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Laboratory-7: Tasks

Submit at https://forms.gle/y7nwWkVS7abNupVL6

by 6:50 PM

Task 1

- For the Steel Plates Faults dataset available at:

```
pip install ucimlrepo

from ucimlrepo import fetch_ucirepo
steel = fetch_ucirepo(id=198)
X = steel.data.features
y = steel.data.targets
```

- Use a 70:30 training-test split and standardise the input features.
- Train a Gaussian Naive Bayes classifiers using the following two cases:
  - Model 1: Equal prior probabilities for all seven classes
  - Model 2: Class prior probabilities estimated from the training dataset (total number of samples in class k/total number of samples)
- Compare the training and test accuracy, precision, recall, and F1-score for both models.
- Identify the best-performing Naive Bayes model and plot its confusion matrix showing all seven classes.
- Use SVM for classifying the same dataset and compare with accuracy of Naive Bayes. Use both One-vs-One and One-vs-Rest.

Task 2

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- For the machine predictive maintenance dataset available at: https://archive.ics.uci.edu/ml/machine-learning-databases/00601/ai4i2020.csv
- Also available at ucimlrepo as data = fetch_ucirepo(id=601)
- Use 5 input features to create a 70:30 training-test split.
- Compare Linear kernel, polynomial kernel and RBF kernel classifiers for the following values of the regularisation parameter: [C = 0.1,1,10,100]
- Compare the training and test accuracy, precision, recall, and F1-score for the different models.
- Identify the best SVM model based on the F1-score performance and plot the confusion matrix.
- Train a Gaussian Naive Bayes classifier using the same training and test dataset. Compare its accuracy, precision, recall, and F1-score with the best SVM model.